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Supplementary Material for "The Guarantee Holds and the Task Fails: Absorbing Stalls of Learning-Enabled Controllers under CBF Filters"

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1. The experiments and their registered verdicts

Table S1: The ten pre-registered experiments of the main text, numbered in reading order, with their registered verdicts; Table S5 maps them to the released records. Evaders: G greedy, H wall-hugging, L learned attacker. Every filtered evaluation had zero collision steps.

	Question	Policies	Evaders	Registered verdict
1	cost of the filter	FREE; 3 remedies	G	filter cost confirmed; filter features recover part, penalty alone none
2	geometry of stalled states	BASE ×3, AWARE	L	98.1% projection agreement (bar 99%: missed); dwell test failed
3	who commands the stall	4 families, FREE, AWARE	L, H, G	3/3 predictions pass; rule D-2
4	is there a signal to escape	AWARE and a fine-tune of it	L, G	"no signal" fails: signal present
5	five learned remedies	BASE, OBS, HOR, penalty, recurrent	L, H, G	primary (immobility) fails on every arm
6	recovery on four families	4 families	L, H, G	recovery passes on 3/3 families
7	recovery parameter N; random baseline	BASE ×3, FREE, AWARE	L, G	capture gain at every N; bound bar missed on 2/10; baseline did not clear deadlock
8	second filter	BASE ×3, AWARE	L, G	stall present on 2/4 (bar 3/4); composition 4/4
9	correction-magnitude penalty	3 reward variants	G	cross-over fails; penalty lowers capture
10	law replaced by a planner	scripted search	G, H, L	law stalls removed 3/3; capture gain 1/3

2. Training details

Observation. The head's base observation has 11 components: own position $x/5$, $y/5$ and heading θ/π; the evader's relative position in the body frame divided by 10 and relative heading divided by π, all zero when the evader is not visible; a visibility flag; and the centres of the two obstacles in the body frame divided by 10. Obstacle radii are not observed; walls are seen only through the own position. The three filter features are the intervention flag of the last executed step, the minimum clearance clipped to $[-0.5, 2]$ m, and the stalled-step run divided by 40 and clipped to $[0, 2]$ (constant after 80 stalled steps). The two allowed-turn features of OBS and OBSHOR are the admissible turn-rate range at zero speed at the current pose. Widths: FREE 11; BASE, HOR, AWARE 14; OBS, OBSHOR 16.

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Reward and discount. $+1$ on capture and -1 on timeout; policies trained behind the filter pay 10^{-3} per stalled step inside a learner transition (the penalty family of Experiment 5 pays 10^{-2}). One learner transition is up to 10 held head steps followed by the intercept law's run until it hands back; in training a transition lasted 14.0–16.6 simulation steps on average (for example 15.9 for BASE seed 1). The discount applies per transition.

PPO. Stable-Baselines3 2.7.1 with its defaults except the discount: MLP policy with two hidden layers of 64 tanh units, 2 048 steps per rollout, minibatch 64, 10 epochs, learning rate 3×10^{-4}, GAE $\lambda = 0.95$, clip 0.2, entropy coefficient 0. The recurrent arm of Experiment 5 used RecurrentPPO with an LSTM policy. The trained families ran 5 M simulation steps, FREE and AWARE 10 M. FREE is the best of five seeds on its confirmation evaluation (capture 0.738 there); AWARE is the best of the three seeds of the remedy with both filter features and the stall penalty.

Intercept law. Chase: full speed with $\omega = \mathrm{clip}(3\,\beta)$ for bearing β; go to the remembered point at full speed until within $0.5\,\mathrm{m}$. The head's hold ends early when the evader becomes visible.

3. Stall classification per policy (Experiment 2)

Table S2: Experiment 2, deterministic actions: shares of the n stalled states. Trap: no admissible command moves the body at the current heading. Position trap: no heading at the position admits motion. Rotation: an admissible rotation with $|\omega| \geq 0.5\,\omega_{\max}$ exists. QP: the exact weighted projection agrees with the filter. Law: the state arose on the intercept-law branch. Self-stop: share of immobile steps on which the requested speed itself was zero.

Policy	n	Exact tier	Trap	Position trap	Rotation	QP	Law	Self-stop
BASE seed 1	74	1.00	1.00	0.00	1.00	1.00	1.00	0.41
BASE seed 2	39	1.00	0.97	0.00	1.00	0.97	0.97	0.66
BASE seed 3	42	1.00	1.00	0.00	1.00	1.00	1.00	0.57
AWARE	107	1.00	0.96	0.00	1.00	0.96	0.57	0.00

4. Penalties and the planner (Experiments 5, 9, 10)

Experiment 9 by reward variant. The registered cross-over hypothesis was that the correction penalty raises capture under the dense varying reward and lowers it under the sparse reward. It failed: the penalty never raised pooled capture; it lowered it significantly only under the dense varying reward. Capture against the greedy evader without $\rightarrow$ with the penalty, seeds 70/71/72, and the pooled paired p:

Table S3: Experiment 9, capture without $\rightarrow$ with the correction penalty.

Reward variant	Seed 70	Seed 71	Seed 72	Pooled p
sparse	0.516 $\rightarrow$ 0.596	0.622 $\rightarrow$ 0.592	0.466 $\rightarrow$ 0.416	1.0
dense, varying (main text)	0.442 $\rightarrow$ 0.338	0.480 $\rightarrow$ 0.300	0.528 $\rightarrow$ 0.212	3.6×10^{-51}
dense, rare	0.620 $\rightarrow$ 0.560	0.508 $\rightarrow$ 0.640	0.538 $\rightarrow$ 0.406	0.13

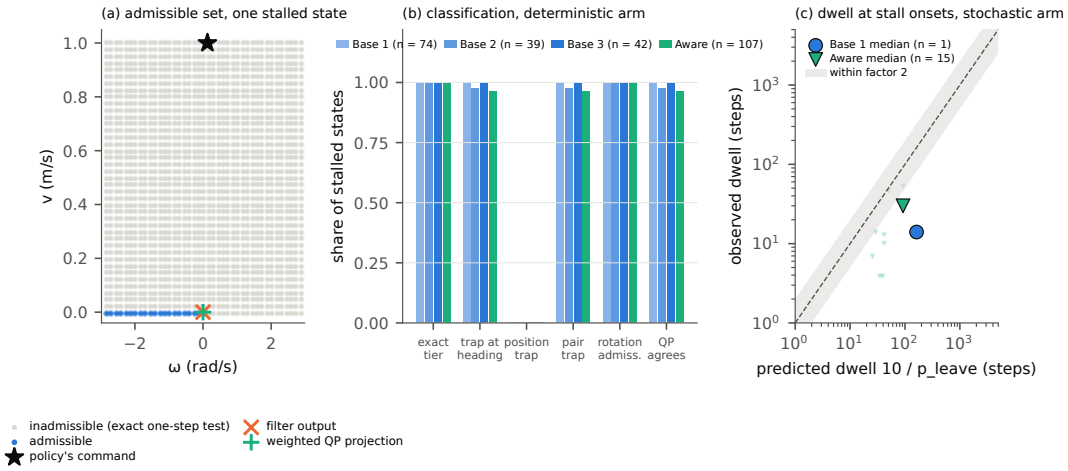


Figure S1: Experiment 2. (a) Admissible set at one stalled state with the policy's command, the filter's output and the exact projection. (b) Classification of stalled states, deterministic actions. (c) Observed against predicted dwell with sampled actions; the registered prediction failed.

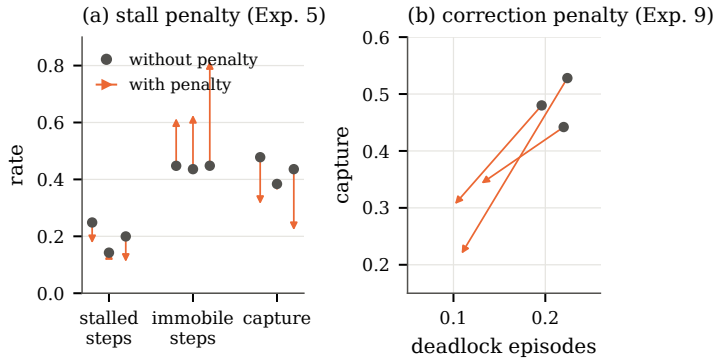


Figure S2: (a) Experiment 5: a stall penalty lowers the stalled-step rate, raises the immobile-step rate and lowers capture, on three seeds. (b) Experiment 9: a correction-magnitude penalty lowers deadlock and capture, on three seeds (dense-reward variant).

Experiment 10. A fully scripted pursuer (scripted orbit search; no learned part) with the line intercept law or an obstacle-avoiding planner law, behind the CBF filter, 500 paired episodes per evader. Registered verdicts: the planner raises capture on at least two of three evaders, FAIL (one of three); law stalls of at least 40 steps fall by at least half, PASS (three of three). Per evader: law-stalled episodes under the line law; how many of them the planner turned into captures; total captures line $\rightarrow$ planner; the paired p:

Table S4: Experiment 10, what became of the law-stalled episodes.

Evader	Law-stalled (line)	Captured (line)	Captured (planner)	Total captures	p
greedy	85	0	16	190 → 203	0.015
wall-hugging	66	8	3	156 → 147	0.035
learned attacker	184	7	0	57 → 46	0.007

5. Recovery details (Experiments 6 and 7)

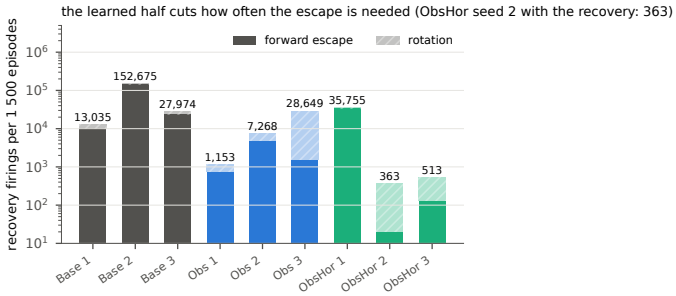


Figure S3: Experiment 6: recovery firings per family over 1 500 evaluation episodes, forward versus turning candidates.

6. Names, records and the first draft's numbering

The experiments were run and archived under internal identifiers, which the released records, checkpoints and pre-registrations keep. Table S5 maps the paper's numbering to them and to the numbering of the first draft.

Table S5: Experiments, earlier numbering, internal round identifiers, seed streams and records (paths under `results/derisk/`).

Exp.	Draft 1	Round	Seed stream	Records
1	1	S4, S5	1 090 000+	`s4_cell_*.json`, `s5_arm_*.json`
2	7	S26 B1	3 000 000+	`s26_b1_round.json`, `s26_b1_cells/`
3	–	S27 (N1)	1 740 000+	`s27_n1_round.json`, `s27_n1_cells/`
4	2	S15	1 710 000+	`s15_round.json`, `s15_cells/`
5	3	S16	1 720 000+	`s16_round.json`, `s16_arm_*.json`
6	4	S17	1 740 000+	`s17_round.json`, `s17_arm_*.json`
7	8	S26 B2	1 740 000+	`s26_b2_round.json`, `s26_b2_cells/`
8	9	S26 B3, B4	1 740 000+	`s26_b2_cells/` (`PRED-*`)
9	5	S18	1 900 000+	`s18_round.json`, `s18_arm_*.json`
10	6	S19	2 000 000+	`s19_round.json`, `s19_cells/`

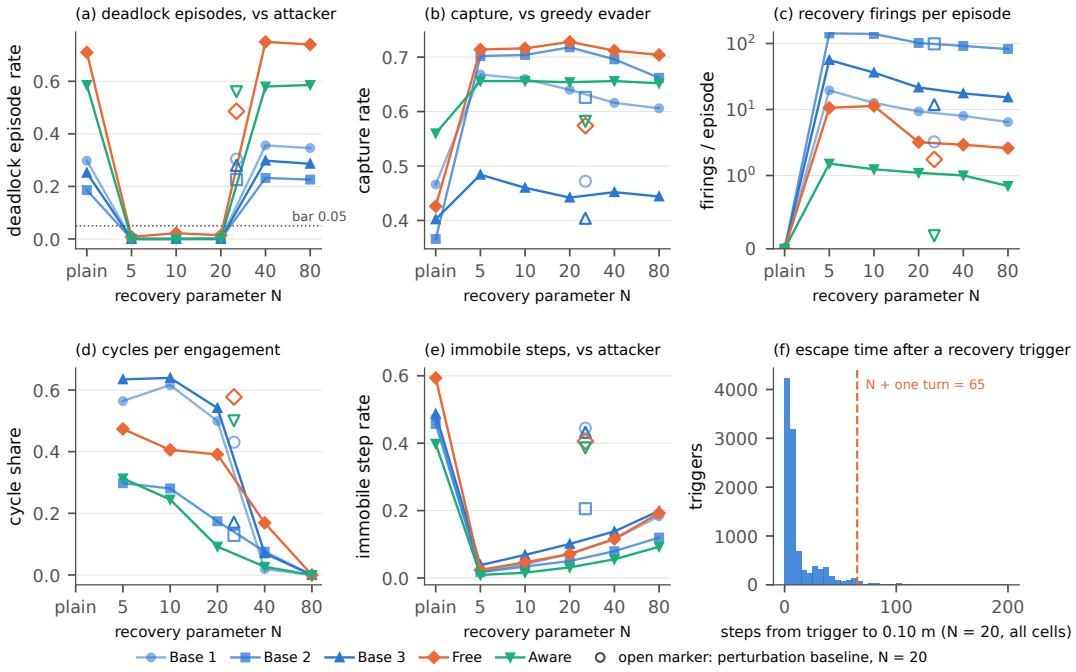


Figure S4: Experiment 7, five policies, 500 paired episodes, against the trigger N: (a) deadlock (attacker; the 40-step count is uninformative for $N \geq 40$ because it includes the recovery's own wait), (b) capture (greedy evader), (c) firings per episode, (d) cycles per engagement, (e) immobile-step rate (attacker); open markers are the random-perturbation baseline at $N = 20$. (f) Steps from trigger to a $0.10\,\mathrm{m}$ displacement at $N = 20$, with the registered bound $N + 45$. The figure counts escaped triggers only (10 466); the record's 10 475 also counts 9 triggers censored with at least 65 steps left, as failures.

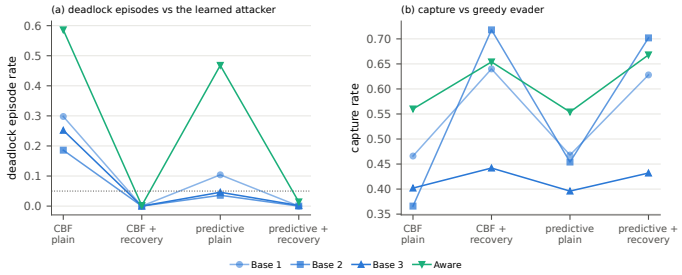


Figure S5: Experiment 8: (a) deadlock against the attacker and (b) capture of the greedy evader under the CBF and predictive filters, without and with the recovery at $N = 20$.

Table S6: Policies, remedies and evaders.

Name in the paper	Repository identifier	Experiments
BASE seeds 1, 2, 3	`BASE-60`, `BASE-61`, `BASE-62`	2, 3, 5, 6, 7, 8
OBS seeds 1, 2, 3	`OBS-60/61/62`	3, 5, 6
HOR seeds 1, 2, 3	`GAM-60/61/62`	3, 5, 6
OBSHOR seeds 1, 2, 3	`OBSGAM-60/61/62`	3, 6
penalty family; recurrent family	`PEN-60/61/62`; `MEM-60/61/62`	5
rotation-only recovery	`REC`	5
recovery with trigger N	`RECF-`N	6, 7, 8
random-perturbation baseline	`PERT-20`	7
predictive filter	`PRED-PLAIN`, `PRED-RECF-20`	8
FREE	`STR14`	1, 3, 7
AWARE; its fine-tune	`RC19`; `FT-B-41`	1–4, 7, 8; 4
remedies of Experiment 1	`RA`, `RB`, `RC` (seeds 19–21)	1
reward variants of Experiment 9	`SPARSE`, `DVARY`, `DRARE` (`-BASE`, `-PEN`, seeds 70–72)	9
line law, planner law	`LINE`, `PLAN`	10
greedy evader; wall-hugging evader; learned attacker	`GREEDY`; `HUG7`; `ATK4-45`	–